KARUNGA JUNIOR SCHOOL

GRADE 9 MATHEMATICS

DETAILED FULL-YEAR LEARNER NOTES

Terms 1, 2 and 3 | Explanations • Worked examples • Practice • Answers

Learner's Name: _____ Class: _____ Year: _____

About this book

This is an expanded teacher-prepared learning resource for Grade 9 Mathematics. It is designed to support classroom teaching, independent study, revision and practice throughout the year. Each topic explains important ideas step by step, gives worked examples, highlights common errors and provides exercises for learners.

These notes follow the topics and sub-strands in the school's Grade 9 annual scheme of work. They are not an official KICD publication. The teacher should guide the pace of learning and confirm any topic-specific requirements against the current KICD curriculum design.

How to study Mathematics

- Read the explanation and understand the meaning of each mathematical term before memorising a formula.
- Copy worked examples into your notebook and explain why each operation is used.
- Show every step in calculations. Do not write only the final answer.
- Draw diagrams neatly and label all known lengths, angles, axes and units.
- Check whether an answer is reasonable. Include units in measurement questions.
- Attempt practice questions independently before checking the answers.
- Ask your teacher to explain any step you cannot justify.

Contents

- Strand 1: Numbers — Integers; Cubes and Cube Roots; Indices and Logarithms; Compound Proportions and Rates of Work
- Strand 2: Algebra — Matrices; Straight Line; Linear Inequalities
- Strand 3: Measurements — Area; Volume of Solids; Mass, Volume, Weight and Density; Time, Distance and Speed; Money; Approximations and Errors
- Strand 4: Geometry — Coordinates and Graphs; Scale Drawing; Similarity and Enlargement; Trigonometry
- Strand 5: Data Handling and Probability — Grouped Data; Probability
- Formula summary and end-of-year revision

STRAND 1: NUMBERS

1.1 INTEGERS

Integers are whole numbers that may be positive, negative or zero. They do not include fractions or decimals. Examples are −12, −3, 0, 5 and 27. Integers are useful when describing temperatures below zero, money owed, bank deposits and withdrawals, elevations above or below sea level, and gains or losses.

A. The number line and absolute value

A number line is a straight line on which numbers are placed in order. Values increase as we move to the right and decrease as we move to the left. Zero separates positive integers from negative integers.

- The opposite of +8 is −8. Both are 8 units from zero.
- The absolute value of a number is its distance from zero. It is written using vertical bars: |−8| = 8 and |8| = 8.
- A negative number with a larger absolute value is smaller. For example, −9 < −4.

Worked example: Compare −7 and −2.

Both are negative. On a number line, −7 lies further to the left than −2.

Therefore, −7 < −2.

B. Addition of integers

When adding integers with the same sign, add their absolute values and retain the common sign. When adding integers with different signs, subtract the smaller absolute value from the larger absolute value and use the sign of the number with the larger absolute value.

Worked example: Evaluate (i) −8 + (−5) and (ii) −13 + 9.

(i) Both numbers are negative. Add 8 + 5 = 13 and retain the negative sign. Answer: −13.

(ii) The signs differ. Subtract 9 from 13 to get 4. Since 13 has the larger absolute value and belongs to −13, the answer is −4.

C. Subtraction of integers

Subtraction can be changed into addition by adding the opposite of the number being subtracted. Remember: keep the first number, change subtraction to addition, and change the sign of the second number.

Worked example: Evaluate 6 − (−4) and −7 − 5.

6 − (−4) = 6 + 4 = 10.

−7 − 5 = −7 + (−5) = −12.

D. Multiplication and division

The sign rules for multiplication and division are the same.

First sign	Second sign	Answer sign
+	+	+
–	–	+
+	–	–
–	+	–

For multiplication, multiply the absolute values and then apply the sign. For division, divide the absolute values and apply the sign. Division by zero is not defined.

Worked example: Evaluate (−7) × 6 and (−84) ÷ (−12).

(−7) × 6: unlike signs give a negative answer; 7 × 6 = 42. Answer: −42.

(−84) ÷ (−12): like signs give a positive answer; 84 ÷ 12 = 7. Answer: 7.

E. Order of operations

When an expression contains several operations, follow this order: brackets; powers or roots; multiplication and division from left to right; addition and subtraction from left to right. This is often remembered as BODMAS.

Worked example: Evaluate $18 - 2 \times (-3) + 4$.

First multiply: $2 \times (-3) = -6$.

The expression becomes $18 - (-6) + 4$.

Subtracting a negative is addition: $18 + 6 + 4 = 28$.

Common mistakes

- Do not assume that two negative signs always give a negative answer. In multiplication, $(-) \times (-)$ is positive.
- Do not confuse -3^2 with $(-3)^2$. The first is -9; the second is 9.
- Do not omit brackets when substituting negative values.

Practice questions

1. Arrange -4, 3, -9, 0 and 7 in ascending order.

2. Evaluate $-15 + 8$.

3. Evaluate $12 - (-9)$.

4. Evaluate $(-6)(-7)$.

5. Evaluate $96 \div (-12)$.

6. Evaluate $-4 + 3 \times (-5)$.

7. A bank account has KSh 350. A withdrawal of KSh 500 is made. Represent the balance as an integer.

Answers for self-check

1. $-9, -4, 0, 3, 7$.

2. -7.

3. 21.

4. 42.

5. -8.

6. -19.

7. -150 (overdrawn by KSh 150).

1.2 CUBES AND CUBE ROOTS

A cube is obtained by multiplying a number by itself three times. The cube of a number a is written a^3. The cube root is the inverse operation: it asks which number, when cubed, gives the stated value.

A. Cubing integers

$a^3 = a \times a \times a$. A positive number cubed is positive. A negative number cubed is negative because there are three negative factors.

Worked example: Find 4^3 and $(-5)^3$.

$4^3 = 4 \times 4 \times 4 = 64$.

$(-5)^3 = (-5) \times (-5) \times (-5) = 25 \times (-5) = -125$.

B. Perfect cubes

A perfect cube is an integer that is the cube of another integer. Learners should become familiar with common perfect cubes.

Number	Cube	Number	Cube
1	1	6	216
2	8	7	343
3	27	8	512
4	64	9	729
5	125	10	1000

C. Finding cube roots

The cube root of N is written $\sqrt[3]{N}$. For example, $\sqrt[3]{64} = 4$ because $4^3 = 64$. Negative numbers also have real cube roots: $\sqrt[3]{(-125)} = -5$.

Worked example: Find $\sqrt[3]{1728}$ using prime factorisation.

Write 1728 as a product of prime factors: $1728 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$.

Group equal factors in sets of three: $(2 \times 2 \times 2)(2 \times 2 \times 2)(3 \times 3 \times 3)$.

Take one factor from each group: $2 \times 2 \times 3 = 12$.

Therefore, $\sqrt[3]{1728} = 12$.

D. Estimating cube roots

If a number is not an obvious perfect cube, identify two consecutive perfect cubes between which it lies. For example, $4^3 = 64$ and $5^3 = 125$. Therefore, the cube root of 100 lies between 4 and 5.

Worked example: Between which two integers does $\sqrt[3]{300}$ lie?

$6^3 = 216$ and $7^3 = 343$.

Since $216 < 300 < 343$, it follows that $6 < \sqrt[3]{300} < 7$.

Practice questions

1. Find 8^3.

2. Find $(-4)^3$.

3. Find $\sqrt[3]{729}$.

4. Find $\sqrt[3]{(-1000)}$.

5. Use prime factorisation to find $\sqrt[3]{3375}$.

6. Between which two consecutive integers does $\sqrt[3]{500}$ lie?

Answers for self-check

1. 512.

2. −64.

3. 9.

4. −10.

5. 15.

6. Between 7 and 8, because $7^3=343$ and $8^3=512$.

1.3 INDICES AND LOGARITHMS

An index (or exponent) tells us how many times a base is used as a factor. In 5^3, 5 is the base and 3 is the index. Thus $5^3 = 5 \times 5 \times 5 = 125$.

A. Laws of indices

For a non-zero base a and suitable integer indices m and n, the following laws apply:

- Multiplication: $a^m \times a^n = a^{m+n}$. Add the indices when the bases are the same.
- Division: $a^m \div a^n = a^{m-n}$, provided $a \neq 0$. Subtract the indices.
- Power of a power: $(a^m)^n = a^{mn}$. Multiply the indices.
- Zero index: $a^0 = 1$, provided $a \neq 0$.
- Negative index: $a^{-n} = 1/a^n$.
- Power of a product: $(ab)^n = a^n b^n$.
- Fractional index: $a^{1/n}$ represents the nth root of a (where defined).

Worked example: Simplify $3^4 \times 3^2 \div 3^3$.

The bases are the same. Add indices for multiplication and subtract for division: $3^{4+2-3} = 3^3$.

$3^3 = 27$.

Worked example: Evaluate 2^{-3}.

A negative index means take the reciprocal: $2^{-3} = 1/2^3 = 1/8$.

B. Standard form

Standard form (scientific notation) expresses a number as $A \times 10^n$, where $1 \leq A < 10$ and n is an integer. It is useful for very large and very small numbers.

Worked example: Write 0.0000562 in standard form.

Move the decimal point five places to the right to obtain 5.62.

The original number is small, so the power is negative: 5.62×10^{-5}.

Worked example: Write 7.31×10^4 as an ordinary number.

Move the decimal point four places to the right: 73 100.

C. Meaning of logarithms

A logarithm answers the question: to what power must a base be raised to obtain a given number? If $a^x = N$, then $\log_a N = x$. For example, since $2^3 = 8$, $\log_2 8 = 3$.

- The base of a logarithm must be positive and cannot equal 1.
- The number whose logarithm is taken must be positive.

- Common logarithms use base 10 and are written log N.
- The logarithm laws are: $\log_a(MN)=\log_a M+\log_a N$; $\log_a(M/N)=\log_a M-\log_a N$; $\log_a(M^n)=n\log_a M$.

Worked example: Evaluate $\log_3 81$.

Ask: 3 to what power gives 81?

Since $3^4 = 81$, $\log_3 81 = 4$.

Worked example: Expand $\log_{10}(100x^2)$.

Using the product law: $\log 100 + \log x^2$.

Using the power law: $2 + 2\log x$, for $x > 0$.

D. Using logarithm tables or a calculator

When a question requires logarithm or antilogarithm tables, follow the table instructions provided by the teacher. A common logarithm has a characteristic and a mantissa. The characteristic gives the power of ten; the mantissa is the decimal part. For numerical work, record sufficient digits and round only at the end unless instructed otherwise.

Practice questions

1. Simplify $4^3 \times 4^5$.

2. Simplify $7^8 \div 7^5$.

3. Evaluate $(2^3)^4$.

4. Evaluate 10^{-2}.

5. Write 6 450 000 in standard form.

6. Write 3.08×10^{-4} as an ordinary decimal.

7. Evaluate $\log_2 64$.

8. Evaluate $\log_{10} 0.01$.

Answers for self-check

1. $4^8 = 65\,536$.

2. $7^3 = 343$.

3. $2^{12} = 4096$.

4. 0.01.

5. 6.45×10^6.

6. 0.000308.

7. 6.

8. −2.

1.4 COMPOUND PROPORTIONS AND RATES OF WORK

A proportion is a statement that two ratios are equal. Proportion is used in recipes, maps, construction, currency calculations, production and work-rate problems.

A. Direct proportion

Two quantities are directly proportional when one increases or decreases in the same ratio as the other. If y is directly proportional to x, y = kx, where k is the constant of proportionality.

Worked example: If 3 exercise books cost KSh 180, find the cost of 8 books.

Cost of one book = 180 ÷ 3 = KSh 60.

Cost of 8 books = 8 × 60 = KSh 480.

B. Inverse proportion

Two quantities are inversely proportional when one increases as the other decreases in such a way that their product remains constant. If y is inversely proportional to x, y = k/x or xy = k.

Worked example: Eight workers complete a job in 15 days. How long would 12 workers take at the same rate?

Workers × days is constant: 8 × 15 = 12 × t.

120 = 12t, so t = 120 ÷ 12 = 10 days.

C. Compound proportion

A quantity may depend on several other quantities. Decide whether each factor is direct or inverse before setting up the calculation. A table is often helpful.

Worked example: Four machines produce 600 items in 5 hours. How many items will 6 identical machines produce in 8 hours?

Output is directly proportional to the number of machines and the time worked.

New output = 600 × (6/4) × (8/5).

= 600 × 1.5 × 1.6 = 1440 items.

D. Rates of work

A rate of work is the fraction of a job completed in one unit of time. If a person completes a job in 8 days, the person's rate is 1/8 of the job per day. When people work together and their rates are independent, add their rates.

Worked example: Pipe A fills a tank in 6 hours and Pipe B fills it in 3 hours. How long do they take together?

Rate of A = 1/6 tank per hour.

Rate of B = 1/3 tank per hour.

Combined rate = 1/6 + 1/3 = 1/2 tank per hour.

Time = 1 ÷ 1/2 = 2 hours.

Important: If a worker or pipe empties rather than fills, its rate is treated as a negative contribution to the net rate.

Practice questions

1. If 5 pens cost KSh 250, find the cost of 12 pens.

2. Six taps fill a tank in 4 hours. How long will 3 identical taps take?

3. Seven workers complete a task in 9 days. How many days will 21 equally efficient workers take?

4. Three machines produce 900 units in 6 hours. How many units will 5 machines produce in 8 hours?

5. A can complete a job in 10 days and B in 15 days. Find their combined time.

Answers for self-check

1. KSh 600.

2. 8 hours.

3. 3 days.

4. 2000 units.

5. 6 days.

STRAND 2: ALGEBRA

2.1 MATRICES

A matrix is a rectangular arrangement of numbers, symbols or expressions in rows and columns. Matrices organise information and are used in data handling, computer graphics, economics and solving systems of equations.

A. Order and elements

The order of a matrix is the number of rows followed by the number of columns. A matrix with 3 rows and 2 columns has order 3 × 2. An element is identified by its row and column position.

Worked example: State the order of A = [[2, 5, 1], [4, 0, −3]].

There are 2 rows and 3 columns.

Therefore, the order is 2 × 3.

B. Types of matrices

- Row matrix: has only one row.
- Column matrix: has only one column.
- Square matrix: has the same number of rows and columns.
- Zero matrix: every element is zero.
- Identity matrix: a square matrix with 1s on the main diagonal and 0s elsewhere.

C. Equality of matrices

Two matrices are equal only when they have the same order and every pair of corresponding elements is equal.

Worked example: Find x and y if [[x, 3], [2, y]] = [[5, 3], [2, −4]].

Compare corresponding entries.

x = 5 and y = −4.

D. Addition and subtraction

Matrices can be added or subtracted only if they have the same order. Add or subtract corresponding elements.

Worked example: Find A + B where A = [[2, 1], [−3, 4]] and B = [[5, −2], [1, 6]].

Add corresponding entries: [[2+5, 1+(−2)], [−3+1, 4+6]].

A + B = [[7, −1], [−2, 10]].

E. Scalar multiplication

A scalar is an ordinary number. To multiply a matrix by a scalar, multiply every element by that number.

Worked example: Find −2[[3, −1], [0, 4]].

Multiply every entry by −2.

Result = [[−6, 2], [0, −8]].

F. Matrix multiplication

Matrix multiplication is not performed by multiplying corresponding entries. Multiply each row of the first matrix by each column of the second matrix and add the products. If A is m × n and B is p × q, AB is defined only when n = p; the product has order m × q.

Worked example: Multiply [[1, 2], [3, 4]] by [[2], [1]].

First row × column: 1×2 + 2×1 = 4.

Second row × column: 3×2 + 4×1 = 10.

The product is the 2 × 1 column matrix [[4], [10]].

Matrix multiplication is generally not commutative: AB is not necessarily equal to BA.

Practice questions

1. State the order of [[1,2],[3,4],[5,6]].

2. Find [[2,3],[1,4]] + [[5,−1],[2,0]].

3. Find 3[[1,−2],[4,0]].

4. Find [[2,1],[0,3]][[1],[4]].

5. Can a 2 × 3 matrix be multiplied by a 2 × 2 matrix on the right? Explain.

Answers for self-check

1. 3 × 2.

2. [[7,2],[3,4]].

3. [[3,−6],[12,0]].

4. [[6],[12]].

5. No. The inner dimensions 3 and 2 are not equal.

2.2 STRAIGHT LINE

A straight line is represented on Cartesian axes by a linear relationship between two variables. The study of straight lines involves coordinates, gradient, intercepts and equations.

A. Cartesian coordinates

The horizontal axis is the x-axis and the vertical axis is the y-axis. Their point of intersection is the origin (0,0). A point is written as an ordered pair (x,y). The x-coordinate is read first and the y-coordinate second.

- Quadrant I: x positive, y positive.
- Quadrant II: x negative, y positive.
- Quadrant III: x negative, y negative.
- Quadrant IV: x positive, y negative.

B. Gradient

Gradient measures the steepness and direction of a line. For two points (x_1,y_1) and (x_2,y_2), $m = (y_2-y_1)/(x_2-x_1)$, provided x_2 is not equal to x_1.

- A positive gradient rises from left to right.
- A negative gradient falls from left to right.
- A horizontal line has gradient zero.
- A vertical line has an undefined gradient because the change in x is zero.

Worked example: Find the gradient of the line through P(2,3) and Q(6,11).

$m = (11-3)/(6-2)$.

$m = 8/4 = 2$.

C. Equation y = mx + c

For a non-vertical straight line, $y = mx + c$, where m is the gradient and c is the y-intercept. The y-intercept is the value of y when x = 0.

Worked example: Find the equation of a line with gradient 3 passing through (2,7).

Use $y = mx + c$. Substitute $m=3$, $x=2$ and $y=7$.

$7 = 3(2) + c$, so $7 = 6 + c$.

$c = 1$. Therefore $y = 3x + 1$.

D. Drawing a straight-line graph

1. Write the equation clearly.
2. Choose convenient values of x.
3. Substitute each x-value into the equation to calculate y.
4. Record the ordered pairs in a table.
5. Choose a suitable scale and label both axes.
6. Plot points accurately and join them using a ruler.

Worked example: Draw y = 2x − 1 for x = −1, 0, 1 and 2.

When $x=-1$, $y=-3$. When $x=0$, $y=-1$. When $x=1$, $y=1$. When $x=2$, $y=3$.

Plot $(-1,-3)$, $(0,-1)$, $(1,1)$, $(2,3)$, then join them with a straight line.

E. Parallel and perpendicular lines

Parallel non-vertical lines have equal gradients. Perpendicular non-vertical lines have gradients whose product is −1. For example, a line with gradient 2 is perpendicular to a line with gradient −1/2.

Practice questions

1. State the quadrant of $(-4,-7)$.

2. Find the gradient through $(1,2)$ and $(5,14)$.

3. Find the y-intercept of $y = -3x + 8$.

4. Find the equation of a line with gradient 4 and y-intercept −2.

5. Find the equation of a line with gradient 2 through $(3,10)$.

6. State the gradient of a line parallel to $y = 5x + 1$.

Answers for self-check

1. III.

2. 3.

3. 8.

4. $y = 4x - 2$.

5. $y = 2x + 4$.

6. 5.

2.3 LINEAR INEQUALITIES

An inequality compares two expressions that are not necessarily equal. It shows that one expression is greater or smaller than another.

A. Inequality symbols

Symbol	Meaning
$<$	less than
$>$	greater than
$\leq$	less than or equal to
$\geq$	greater than or equal to

B. Solving inequalities

The same operation can be performed on both sides of an inequality. However, when multiplying or dividing both sides by a negative number, reverse the inequality sign.

Worked example: Solve $4x - 3 \leq 13$.

Add 3 to both sides: $4x \leq 16$.

Divide both sides by 4: $x \leq 4$.

Worked example: Solve $-5x > 20$.

Divide both sides by −5. Reverse the sign because division is by a negative number.

$x < -4$.

C. Representing solutions on a number line

- For $x > 3$, use an open circle at 3 and shade to the right.
- For $x \geq 3$, use a closed circle at 3 and shade to the right.
- For $x < 3$, use an open circle at 3 and shade to the left.
- For $x \leq 3$, use a closed circle at 3 and shade to the left.

An open circle means the boundary value is not included. A closed circle means the boundary value is included.

D. Checking a solution

Choose a value from the proposed solution set and substitute it into the original inequality. The statement should be true.

Worked example: Check whether $x = 2$ satisfies $3x + 1 < 9$.

Substitute $x=2$: $3(2)+1 < 9$.

$7 < 9$ is true. Therefore $x=2$ is a solution.

Practice questions

1. Solve $x + 7 > 12$.

2. Solve $3x - 5 \leq 10$.

3. Solve $-2x \geq 8$.

4. Solve $5 - x < 2$.

5. Represent $x \geq -1$ on a number line.

Answers for self-check

1. $x > 5$.

2. $x \leq 5$.

3. $x \leq -4$.

4. $x > 3$.

5. Closed circle at -1, shading to the right.

STRAND 3: MEASUREMENTS

3.1 AREA

Area is the amount of surface enclosed by a two-dimensional shape. Area is measured in square units, such as mm^2, cm^2 and m^2. Always convert measurements to the same unit before using a formula.

A. Perimeter and area

Perimeter is the total distance around a shape; area measures the surface inside it. Do not confuse the two. Perimeter is in units of length, while area is in square units.

B. Formulae for plane shapes

Shape	Area formula	Important detail
Square	$A = s^2$	s is the side length.
Rectangle	$A = l \times w$	l is length; w is width.
Triangle	$A = \frac{1}{2}bh$	h is perpendicular height.
Parallelogram	$A = bh$	h is perpendicular height.
Trapezium	$A = \frac{1}{2}(a+b)h$	a and b are parallel sides.
Circle	$A = \pi r^2$	r is radius.
Semicircle	$A = \frac{1}{2}\pi r^2$	Half the area of a circle.

C. Circle circumference

Circumference is the distance around a circle. $C = 2\pi r$ or $C = \pi d$, where d is diameter and $d = 2r$. Use the value of π instructed in the question.

Worked example: A circular garden has diameter 14 m. Find its area and circumference using $\pi = 22/7$.

Radius = $14 \div 2 = 7$ m.

Area = $\pi r^2 = 22/7 \times 7 \times 7 = 154$ m².

Circumference = $\pi d = 22/7 \times 14 = 44$ m.

D. Composite shapes

A composite shape is made from two or more familiar shapes. Divide it into rectangles, triangles, circles or other shapes. Calculate each area and add them. If a section is cut out, subtract its area.

Worked example: A shape consists of a rectangle 10 cm by 6 cm with a right-angled triangle of base 4 cm and height 6 cm attached. Find total area.

Rectangle area = $10 \times 6 = 60$ cm².

Triangle area = $\frac{1}{2} \times 4 \times 6 = 12$ cm².

Total area = $60 + 12 = 72$ cm².

E. Unit conversions

1 m = 100 cm, but 1 m² = 10 000 cm². When converting area units, square the linear conversion factor. For example, 1 m² = 100 cm × 100 cm = 10 000 cm².

Practice questions

1. Find the area of a square of side 11 cm.

2. Find the area of a triangle with base 16 cm and perpendicular height 9 cm.

3. Find the area of a trapezium with parallel sides 8 cm and 14 cm and height 5 cm.

4. Find the circumference of a circle of radius 7 cm using $\pi=22/7$.

5. A rectangular floor is 8 m by 5 m. Find its area in cm².

6. A circle has diameter 10 cm. Find its area in terms of π.

Answers for self-check

1. 121 cm².

2. 72 cm².

3. 55 cm^2.

4. 44 cm.

5. 400 000 cm^2.

6. 25π cm^2.

3.2 VOLUME OF SOLIDS

Volume is the amount of three-dimensional space occupied by an object. It is measured in cubic units, such as cm^3 and m^3. Capacity is the amount a container can hold and may be measured in litres or millilitres.

A. Volume of a cuboid and cube

Cuboid: V = length × width × height. Cube: V = side × side × side = s^3. The dimensions must be in the same unit.

Worked example: Find the volume of a cuboid measuring 12 cm by 5 cm by 4 cm.

V = lwh = 12 × 5 × 4 = 240 cm^3.

B. Volume of a prism

A prism has a uniform cross-section along its length. Volume = area of cross-section × length. The cross-section may be a triangle, rectangle or another polygon.

Worked example: A triangular prism has a triangular cross-section with base 6 cm and height 4 cm. Its length is 10 cm. Find volume.

Area of triangle = ½ × 6 × 4 = 12 cm^2.

Volume = 12 × 10 = 120 cm^3.

C. Cylinder

A cylinder has two parallel circular ends. Its volume is $V = \pi r^2 h$, where r is radius and h is perpendicular height.

Worked example: Find the volume of a cylinder of radius 3.5 cm and height 10 cm, using π=22/7.

$V = \pi r^2 h = 22/7 \times 3.5 \times 3.5 \times 10$.

V = 385 cm^3.

D. Pyramid and cone

A pyramid has a polygonal base and triangular faces meeting at an apex. Its volume is one-third of the product of base area and perpendicular height: $V = \frac{1}{3}Bh$. A cone has a circular base, so $V = \frac{1}{3}\pi r^2 h$.

Worked example: A cone has radius 7 cm and height 12 cm. Find its volume using π=22/7.

$V = \frac{1}{3} \times 22/7 \times 7^2 \times 12$.

$= \frac{1}{3} \times 22 \times 7 \times 12 = 616$ cm^3.

E. Sphere

A sphere is a perfectly round solid. Its volume is $V = \frac{4}{3}\pi r^3$. A hemisphere is half a sphere, so its volume is $\frac{2}{3}\pi r^3$.

F. Capacity and volume conversions

- 1 cm^3 = 1 mL.

- 1000 cm^3 = 1 litre.
- 1 m^3 = 1 000 000 cm^3 = 1000 litres.

Use these relationships when converting the volume of a container into its capacity.

Practice questions

1. Find the volume of a cube of side 6 cm.

2. Find the volume of a cuboid 15 cm × 8 cm × 4 cm.

3. A prism has cross-sectional area 18 cm^2 and length 7 cm. Find its volume.

4. Find the volume of a cylinder of radius 7 cm and height 8 cm using $\pi = 22/7$.

5. Find the volume of a cone of radius 3 cm and height 7 cm using $\pi = 22/7$.

6. Convert 2500 cm^3 to litres.

Answers for self-check

1. 216 cm^3.

2. 480 cm^3.

3. 126 cm^3.

4. 1232 cm^3.

5. 66 cm^3.

6. 2.5 litres.

3.3 MASS, VOLUME, WEIGHT AND DENSITY

Mass, volume, weight and density are related but describe different physical quantities. Correct units and formulae are important.

A. Mass and weight

Mass is the amount of matter in an object. It is measured in kilograms (kg) or grams (g). Weight is the force with which gravity attracts an object and is measured in newtons (N).

Weight W = mg, where m is mass in kilograms and g is gravitational field strength in N/kg. In many school questions, g may be taken as 10 N/kg if stated.

Worked example: Find the weight of a 7 kg object when g = 10 N/kg.

W = mg = 7 × 10 = 70 N.

B. Density

Density tells us how much mass is contained in a unit volume. Density = mass ÷ volume. The symbol commonly used is ρ (rho).

$\rho = m/V$; $m = \rho V$; $V = m/\rho$. Common units include kg/m^3 and g/cm^3.

Worked example: A metal block has mass 540 g and volume 60 cm^3. Find its density.

Density = mass/volume = 540/60 = 9 g/cm^3.

Worked example: An object has density 2.7 g/cm³ and volume 20 cm³. Find its mass.

Mass = density × volume = 2.7 × 20 = 54 g.

C. Floating and sinking

For objects in the same fluid, an object with average density less than the fluid may float; an object with average density greater than the fluid tends to sink. Shape and trapped air can affect the average density of an object.

D. Unit conversion

- 1 kg = 1000 g.
- 1 m³ = 1 000 000 cm³.
- 1 g/cm³ = 1000 kg/m³.

Convert mass and volume into compatible units before dividing.

Practice questions

1. Find density if mass is 480 g and volume is 160 cm³.

2. Find mass if density is 8 g/cm³ and volume is 25 cm³.

3. Find volume if mass is 150 g and density is 3 g/cm³.

4. Find weight of a 12 kg object where g=10 N/kg.

5. Convert 2.5 g/cm³ into kg/m³.

Answers for self-check

1. 3 g/cm³.

2. 200 g.

3. 50 cm³.

4. 120 N.

5. 2500 kg/m³.

3.4 TIME, DISTANCE AND SPEED

Speed describes how quickly an object moves. It is the distance travelled per unit time. Average speed is the total distance divided by the total time taken.

A. Main formulae

Quantity	Formula
Speed	v = d/t
Distance	d = vt
Time	t = d/v

B. Units and conversion

Distance may be measured in metres or kilometres. Time may be measured in seconds, minutes or hours. Speed may be in m/s or km/h. Use consistent units.

- To change km/h to m/s, divide by 3.6.
- To change m/s to km/h, multiply by 3.6.

Worked example: Convert 90 km/h to m/s.

90 ÷ 3.6 = 25 m/s.

C. Average speed and journeys

For a journey with different speeds, add all distances and divide by the total time. Do not simply average the speeds unless equal time intervals or other specified conditions make that valid.

Worked example: A vehicle travels 120 km in 2 hours and then 60 km in 1 hour. Find average speed.

Total distance = 120 + 60 = 180 km.

Total time = 2 + 1 = 3 hours.

Average speed = 180 ÷ 3 = 60 km/h.

D. Time calculations

Convert minutes into hours by dividing by 60. Convert seconds into minutes by dividing by 60. For example, 1 hour 30 minutes = 1.5 hours.

Practice questions

1. A car travels 210 km in 3 hours. Find its average speed.

2. A runner moves at 6 m/s for 45 seconds. Find distance.

3. How long does a bus take to cover 240 km at 80 km/h?

4. Convert 54 km/h to m/s.

5. A cyclist covers 18 km in 45 minutes. Find average speed in km/h.

Answers for self-check

1. 70 km/h.

2. 270 m.

3. 3 hours.

4. 15 m/s.

5. 24 km/h.

3.5 MONEY

Mathematics is used in daily financial decisions such as shopping, budgeting, saving, borrowing, banking and running a business. Accurate calculations help people use money responsibly.

A. Cost price, selling price, profit and loss

Cost price (CP) is the amount paid to obtain an item. Selling price (SP) is the amount received when it is sold.

- Profit = SP − CP, when SP is greater than CP.
- Loss = CP − SP, when CP is greater than SP.
- Percentage profit = profit/CP × 100%.
- Percentage loss = loss/CP × 100%.

Worked example: A trader buys a bag for KSh 1,600 and sells it for KSh 2,000. Find profit and percentage profit.

Profit = 2,000 − 1,600 = KSh 400.

Percentage profit = 400/1,600 × 100 = 25%.

B. Discount and marked price

Marked price is the displayed price before discount. Discount is the reduction in price.

- Discount = discount rate × marked price.
- Selling price after discount = marked price − discount.

Worked example: A phone accessory marked KSh 2,500 is discounted by 12%. Find the sale price.

Discount = 12/100 × 2500 = KSh 300.

Sale price = 2500 − 300 = KSh 2,200.

C. Simple interest

Simple interest is calculated on the original principal for the entire period.

I = PRT/100, where P is principal, R is annual interest rate as a percentage, and T is time in years. Total amount A = P + I.

Worked example: Find simple interest on KSh 8,000 at 6% per annum for 3 years.

I = 8000 × 6 × 3 / 100.

I = KSh 1,440.

Amount = 8,000 + 1,440 = KSh 9,440.

D. Budgeting

A budget is a plan of expected income and expenditure over a period. Income may include wages, sales or allowances. Expenditure includes food, transport, rent, learning materials and savings.

- Surplus = income − expenditure, when income is greater.
- Deficit occurs when expenditure is greater than income.
- A realistic budget includes needs, planned savings and unexpected expenses.

Worked example: A learner's monthly income is KSh 4,000. Expenses are KSh 1,200, KSh 800, KSh 500 and KSh 700. Find the balance.

Total expenses = 1,200 + 800 + 500 + 700 = KSh 3,200.

Balance = 4,000 − 3,200 = KSh 800 surplus.

Practice questions

1. An item costs KSh 950 and is sold for KSh 1,140. Find profit and percentage profit.

2. An item bought for KSh 2,400 is sold at a loss of KSh 360. Find percentage loss.

3. Find 15% discount on KSh 4,800 and the sale price.

4. Find simple interest on KSh 15,000 at 8% for 2 years.

5. A household earns KSh 32,000 and spends KSh 29,500. Find the surplus.

Answers for self-check

1. KSh 190; 20%.

2. 15%.

3. KSh 720; sale price KSh 4,080.

4. KSh 2,400.

5. KSh 2,500.

3.6 APPROXIMATIONS AND ERRORS

An approximation is a value close to the exact value. Approximations are useful when exact values are unnecessary or impossible to obtain. Measurement errors arise because instruments and methods have limitations.

A. Rounding off to decimal places

Decimal places count the digits after the decimal point. To round to a required number of decimal places, identify the next digit. If it is 5, 6, 7, 8 or 9, increase the last retained digit by one. If it is 0, 1, 2, 3 or 4, leave the last retained digit unchanged.

Worked example: Round 37.486 to two decimal places.

Keep two decimal digits: 37.48.

The next digit is 6, so increase 8 by 1.

Answer: 37.49.

B. Significant figures

Significant figures are the meaningful digits in a number. Start counting from the first non-zero digit.

- Leading zeros are not significant: 0.0045 has two significant figures.
- Zeros between non-zero digits are significant: 205 has three significant figures.
- Trailing zeros after a decimal point are significant: 2.500 has four significant figures.

Worked example: Round 0.006784 to three significant figures.

The first three significant digits are 6, 7 and 8.

The next digit is 4, so do not increase the 8.

Answer: 0.00678.

C. Upper and lower bounds

If a measurement is rounded to the nearest unit, the actual value lies within half a unit below and half a unit above the rounded value. For example, a length recorded as 12 cm to the nearest centimetre has lower bound 11.5 cm and upper bound 12.5 cm (the upper bound is excluded under the usual rounding convention).

D. Absolute and percentage error

When a true or accepted value is known, absolute error = |measured value − true value|. Percentage error = (absolute error ÷ true value) × 100%.

Worked example: A length has an accepted value of 25 cm but is measured as 24.5 cm. Find absolute and percentage error.

Absolute error = $|24.5 - 25| = 0.5$ cm.

Percentage error = $0.5/25 \times 100 = 2\%$.

Practice questions

1. Round 56.738 to 1 decimal place.

2. Round 0.000946 to 2 significant figures.

3. Write 23 650 to 3 significant figures.

4. A mass is recorded as 8 kg to the nearest kilogram. State its lower and upper bounds.

5. A measured distance is 48 m while the accepted value is 50 m. Find absolute and percentage error.

Answers for self-check

1. 56.7.

2. 0.00095.

3. 23 700.

4. $7.5 \text{ kg} \leq \text{mass} < 8.5 \text{ kg}$.

5. 2 m; 4%.

STRAND 4: GEOMETRY

4.1 COORDINATES AND GRAPHS

Coordinates are used to locate points on a plane. Graphs represent relationships between variables and help us identify patterns, compare values and make estimates.

A. The Cartesian plane

The x-axis is horizontal and the y-axis is vertical. The origin is (0,0). The plane is divided into four quadrants. To plot (−2,3), move 2 units left from the origin and 3 units up.

B. Tables of values

A table of values helps generate points for a graph. Substitute each chosen x-value into the rule to calculate y.

Worked example: Complete a table for $y = 3x - 2$ when $x = -1, 0, 1$ and 2.

$x=-1$: $y=-3-2=-5$.

$x=0$: $y=-2$.

$x=1$: $y=1$.

$x=2$: $y=4$.

The points are (−1,−5), (0,−2), (1,1) and (2,4).

C. Interpreting graphs

Read the scale on each axis carefully. Check whether the axis begins at zero. A graph may show a direct relationship, an increasing or decreasing trend, or a constant value.

- Interpolation estimates a value within the range of the plotted data.
- Extrapolation estimates beyond the available data and is less reliable.
- The gradient of a straight-line graph is change in vertical value divided by change in horizontal value.

D. Distance-time graphs

On a distance-time graph, time is usually on the horizontal axis and distance on the vertical axis. A straight rising line represents constant speed. A horizontal line represents rest. A steeper line indicates greater speed.

Worked example: A distance-time graph shows a learner covering 300 m in 60 seconds at constant speed. Find the speed.

Speed = gradient = change in distance/change in time.

= 300/60 = 5 m/s.

Practice questions

1. Plot the points (0,2), (1,4), (2,6), (3,8). State the rule.

2. Find the gradient between (2,3) and (6,15).

3. A cyclist covers 600 m in 2 minutes. Find average speed in m/s.

4. Explain what a horizontal section on a distance-time graph means.

Answers for self-check

1. y = 2x + 2.

2. 3.

3. 600/120 = 5 m/s.

4. The object is stationary; distance remains constant as time passes.

4.2 SCALE DRAWING

A scale drawing is a representation of an object or distance using a fixed ratio between drawing measurements and actual measurements. It is used in maps, building plans, engineering drawings and route planning.

A. Understanding scale

A scale of 1:100 means one unit on the drawing represents 100 of the same units in real life. The units must match before multiplying.

Worked example: A room is 6 cm long on a plan drawn to scale 1:50. Find the actual length.

Actual length = 6 × 50 = 300 cm.

300 cm = 3 m.

B. Reducing and enlarging

A scale smaller than 1:1 reduces the size of the drawing. A scale such as 5:1 enlarges the object. Read the order of the ratio carefully.

C. Map scales

A map may use a ratio scale, a statement scale or a linear scale. For a statement scale of 1 cm represents 4 km, a measured map distance of 7 cm represents 28 km.

Worked example: Two towns are 8.5 cm apart on a map. The scale is 1 cm to 6 km. Find actual distance.

Actual distance = 8.5 × 6 = 51 km.

D. Bearings

A bearing is an angle measured clockwise from north. It is written using three digits, for example 045°, 090° and 275°. Draw a north line at the starting point, place the protractor at that point, and measure clockwise.

- North = 000° or 360°.
- East = 090°.
- South = 180°.
- West = 270°.

Practice questions

1. A drawing uses scale 1:200. A length is 7 cm on the drawing. Find actual length in metres.

2. A map scale is 1 cm to 3.5 km. Find actual distance for 9 cm.

3. A rectangular field measures 5 cm by 3 cm on a 1:1000 plan. Find its actual dimensions in metres.

4. State the bearing of south-east.

5. Explain why all units must be made the same before using a ratio scale.

Answers for self-check

1. 14 m.

2. 31.5 km.

3. 50 m by 30 m.

4. 135°.

5. Ratios compare like quantities; mismatched units give an incorrect scale calculation.

4.3 SIMILARITY AND ENLARGEMENT

Similar shapes have the same shape but may have different sizes. Corresponding angles are equal and corresponding sides are in the same ratio.

A. Similarity

To establish that two polygons are similar, check that corresponding angles are equal and corresponding sides are proportional. For triangles, common tests include equal corresponding angles (AAA) or proportional corresponding sides with the included angle equal (SAS similarity).

Worked example: Two similar triangles have corresponding sides 4 cm and 10 cm. Find the scale factor from the smaller to the larger triangle.

Scale factor = image length/original length = 10/4 = 2.5.

B. Enlargement

An enlargement changes the size of a shape from a fixed centre. A positive scale factor greater than 1 enlarges; a positive factor between 0 and 1 reduces. A negative scale factor places the image on the opposite side of the centre.

To enlarge a point from centre O by scale factor k, draw a line from O through the point and mark the image at k times the original distance from O.

C. Length, area and volume scale factors

- If the length scale factor is k, the area scale factor is k^2.
- If the length scale factor is k, the volume scale factor is k^3.

Worked example: Two similar solids have corresponding lengths in the ratio 2:3. Find the ratio of their volumes.

Volume ratio = $2^3:3^3$ = 8:27.

Worked example: A shape is enlarged by scale factor 4. Its original area is 6 cm^2. Find the enlarged area.

Area scale factor = 4^2 = 16.

Enlarged area = 6 × 16 = 96 cm^2.

D. Common errors

- Do not use the length scale factor directly for area or volume.
- Match corresponding sides in the same order.
- Use the centre of enlargement consistently when drawing an image.

Practice questions

1. A 5 cm side is enlarged by scale factor 2.4. Find image length.

2. Find the area ratio for similar shapes with length ratio 3:5.

3. Find the volume ratio for similar solids with length ratio 2:5.

4. A figure is reduced by scale factor 1/3. A side is 12 cm. Find its image length.

5. A shape is enlarged by factor 3. Its original area is 7 cm^2. Find new area.

Answers for self-check

1. 12 cm.

2. 9:25.

3. 8:125.

4. 4 cm.

5. 63 cm^2.

4.4 TRIGONOMETRY

Trigonometry studies relationships between angles and sides of triangles. In Grade 9, right-angled triangles are especially important.

A. Naming the sides

A right-angled triangle has one angle of 90°. The hypotenuse is the longest side and lies opposite the right angle. For a chosen acute angle θ, the opposite side is directly across from θ and the adjacent side is next to θ but is not the hypotenuse.

B. The three ratios

Ratio	Formula	Remember
Sine	sin θ = opposite/hypotenuse	SOH
Cosine	cos θ = adjacent/hypotenuse	CAH
Tangent	tan θ = opposite/adjacent	TOA

The memory aid SOH-CAH-TOA helps learners select the correct ratio.

C. Finding a side

Worked example: A right triangle has hypotenuse 10 cm and angle 30°. Find the opposite side.

Use sine because opposite and hypotenuse are involved.

sin 30° = opposite/10.

0.5 = opposite/10.

Opposite = 10 × 0.5 = 5 cm.

Worked example: A right triangle has adjacent side 8 cm and angle 60°. Find the hypotenuse.

Use cosine: cos 60° = adjacent/hypotenuse.

0.5 = 8/hypotenuse.

Hypotenuse = 8 ÷ 0.5 = 16 cm.

D. Finding an angle

Use inverse trigonometric functions when two sides are known and the angle is required. Ensure the calculator is in degree mode if the question uses degrees.

Worked example: A right triangle has opposite side 5 cm and adjacent side 5 cm. Find θ.

tan θ = 5/5 = 1.

θ = $\tan^{-1}(1)$ = 45°.

E. Applications

Trigonometry can be used to estimate the height of a building, the length of a ramp, the angle of elevation to an object and the distance across an inaccessible area. Draw and label a right-angled triangle before choosing a ratio.

Angle of elevation is measured upwards from a horizontal line. Angle of depression is measured downwards from a horizontal line. When solving a word problem, identify the horizontal and vertical distances carefully.

F. Calculator checks

- Check that the calculator is set to degrees when angles are in degrees.

- Use brackets where necessary.
- Keep sufficient decimal places during working and round the final answer as instructed.
- Check that a hypotenuse is longer than either shorter side.

Practice questions

1. Find opposite side when hypotenuse is 14 cm and $\theta=30°$.

2. Find adjacent side when hypotenuse is 12 cm and $\theta=60°$.

3. Find θ when opposite=8 cm and adjacent=8 cm.

4. A ladder 5 m long makes an angle of 60° with the ground. Find the vertical height reached.

5. State the ratio that uses opposite and adjacent sides.

Answers for self-check

1. 7 cm.

2. 6 cm.

3. 45°.

4. $5 \sin 60° \approx 4.33$ m.

5. Tangent.

STRAND 5: DATA HANDLING AND PROBABILITY

5.1 GROUPED DATA

Data are facts or measurements collected for a purpose. When a data set is large, values may be grouped into class intervals. Grouping makes data easier to organise and interpret, but some exact information is lost.

A. Frequency tables

A frequency table lists class intervals and the number of observations in each interval. Frequency is represented by f. The sum of all frequencies is the total number of observations, $N = \Sigma f$.

B. Class boundaries and midpoints

For continuous grouped data, class boundaries may be used to remove gaps between classes. The class midpoint is the average of the lower and upper class boundaries: midpoint x = (lower boundary + upper boundary)/2.

Worked example: Find the midpoints of the intervals 0–10, 10–20 and 20–30.

Midpoint of 0–10 = (0+10)/2 = 5.

Midpoint of 10–20 = 15.

Midpoint of 20–30 = 25.

C. Estimated mean

The exact mean of grouped data cannot usually be found because individual observations are not shown. An estimate is calculated using class midpoints.

Estimated mean = Σfx/Σf, where f is frequency and x is class midpoint.

Worked example: Find the estimated mean for the table below.

Class intervals: 0–10, 10–20, 20–30. Frequencies: 2, 3, 5.

Midpoints x: 5, 15, 25.

fx values: 10, 45, 125. Therefore Σfx = 180 and Σf = 10.

Estimated mean = 180/10 = 18.

D. Median and cumulative frequency

Cumulative frequency is the running total of frequencies. It helps locate the median class. For N observations, the median position is around N/2. Identify the class interval whose cumulative frequency first reaches or exceeds that position.

E. Modal class and histogram

The modal class is the class interval with the highest frequency. A histogram represents grouped continuous data using adjoining rectangles. When class widths differ, frequency density (frequency ÷ class width) is used for the vertical axis.

Practice questions

1. Find the midpoint of 30–50.

2. Frequencies are 3, 5, 7 and 5. Find total frequency.

3. For classes 0–10, 10–20 and 20–30 with frequencies 1, 4 and 5, calculate estimated mean.

4. Which class is modal if frequencies are 4, 9, 6 and 3?

5. Explain one limitation of estimating the mean from grouped data.

Answers for self-check

1. 40.

2. 20.

3. 20.

4. The second class (frequency 9).

5. Individual data values are not known; class midpoints are used, so the mean is an estimate.

5.2 PROBABILITY

Probability is a measure of the chance that an event will occur. It is used in games, weather forecasts, decision-making and many everyday situations.

A. Outcomes, sample space and events

An outcome is a possible result of an experiment. The sample space is the set of all possible outcomes. An event is a collection of outcomes of interest.

Worked example: A fair coin is tossed once. State the sample space and event of getting a head.

Sample space S = {H,T}.

Event A = {H}.

B. Probability of equally likely outcomes

When all outcomes are equally likely, P(A) = number of favourable outcomes ÷ total number of possible outcomes. Probability is between 0 and 1 inclusive.

Worked example: A fair die is rolled. Find the probability of getting a prime number.

Sample space = {1,2,3,4,5,6}.

Prime numbers are 2, 3 and 5: three favourable outcomes.

P(prime) = 3/6 = 1/2.

C. Complementary events

The complement of event A is the event that A does not happen. P(not A) = 1 − P(A).

Worked example: The probability of rain is 0.3. Find the probability that it does not rain.

P(no rain) = 1 − 0.3 = 0.7.

D. Mutually exclusive events

Two events are mutually exclusive if they cannot happen at the same time. For mutually exclusive events A and B, P(A or B) = P(A) + P(B).

E. Experimental probability

Experimental probability is based on actual results from repeated trials: number of times the event occurs ÷ total number of trials. Experimental results may differ from theoretical probability, but often approach it as the number of trials becomes large.

Worked example: A coin is tossed 50 times and heads occurs 28 times. Find experimental probability of heads.

P(heads) = 28/50 = 14/25 = 0.56.

F. Two-stage experiments

For two independent events, the probability of both occurring is the product of their probabilities. For example, the probability of getting two heads in two fair coin tosses is 1/2 × 1/2 = 1/4. A tree diagram can help organise outcomes.

Practice questions

1. A fair die is rolled. Find P(rolling a 6).

2. A bag contains 4 red, 3 blue and 5 green balls. Find P(blue).

3. If P(A)=0.62, find P(not A).

4. A coin is tossed twice. List the sample space.

5. A fair coin is tossed twice. Find probability of two tails.

6. In 40 trials an event occurs 15 times. Find its experimental probability.

Answers for self-check

1. 1/6.

2. 3/12 = 1/4.

3. 0.38.

4. {HH, HT, TH, TT}.

5. 1/4.

6. 15/40 = 3/8 = 0.375.

FORMULA SUMMARY

Topic	Formula / rule
Integers	Use sign rules; subtract by adding the opposite.
Cube	$a^3 = a \times a \times a$; cube root $\sqrt[3]{a}$ reverses cubing.
Indices	$a^m a^n = a^{m+n}$; $a^m/a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$; $a^{-n} = 1/a^n$.
Direct proportion	$y = kx$.
Inverse proportion	$y = k/x$ or $xy = k$.
Rate of work	Rate = fraction of job/time; combined independent rates are added.
Matrices	For AB, columns of A = rows of B.
Straight line	$y = mx + c$; $m = (y_2 - y_1)/(x_2 - x_1)$.
Rectangle	$A = lw$.
Triangle	$A = \frac{1}{2}bh$.
Trapezium	$A = \frac{1}{2}(a+b)h$.
Circle	Area $= \pi r^2$; circumference $= 2\pi r = \pi d$.
Cuboid	$V = lwh$.
Prism	V = cross-sectional area × length.
Cylinder	$V = \pi r^2 h$.
Cone	$V = \frac{1}{3}\pi r^2 h$.
Pyramid	$V = \frac{1}{3} \times$ base area × perpendicular height.
Sphere	$V = \frac{4}{3}\pi r^3$.
Density	$\rho = m/V$; $m = \rho V$; $V = m/\rho$.
Weight	$W = mg$.
Speed	$v = d/t$; $d = vt$; $t = d/v$.
Simple interest	$I = PRT/100$; amount $= P + I$.
Percentage error	absolute error/true value × 100%.
Scale	Actual length = drawing length × scale factor for 1:n.
Similarity	Length ratio k; area ratio k^2; volume ratio k^3.
Trigonometry	$\sin\theta = O/H$; $\cos\theta = A/H$; $\tan\theta = O/A$.
Grouped mean	Estimated mean $= \Sigma fx/\Sigma f$.
Probability	P(A) = favourable outcomes/total equally likely outcomes; P(not A) = 1 − P(A).

END-OF-YEAR REVISION TASK

Attempt these questions in your exercise book. Show full working and include units where appropriate.

1. Evaluate $-18 + 7 - (-5)$.

2. Find $\sqrt[3]{2744}$.

3. Simplify $5^4 \times 5^2 \div 5^3$.

4. Write 0.000083 in standard form.

5. Ten workers finish a task in 12 days. How many days would 15 equally efficient workers take?

6. Add the matrices [[2,3],[1,4]] and [[−1,5],[3,0]].

7. Find the equation of a line with gradient 3 passing through (2,8).

8. Solve $7 - 2x \geq 15$.

9. Find the area of a circle of radius 14 cm using $\pi = 22/7$.

10. Find the volume of a cylinder of radius 3 cm and height 7 cm in terms of π.

11. A block has mass 900 g and volume 300 cm^3. Find density.

12. A vehicle travels 180 km in 2.5 hours. Find average speed.

13. Find simple interest on KSh 20,000 at 7% per annum for 2 years.

14. Round 0.008764 to 2 significant figures.

15. Find the gradient between (−1,2) and (3,14).

16. A map scale is 1 cm to 8 km. What actual distance is represented by 4.5 cm?

17. Similar shapes have length ratio 3:4. Find area ratio.

18. Find the opposite side of a right triangle with hypotenuse 20 cm and angle 30°.

19. Find the estimated mean for midpoints 5,15,25 with frequencies 2,4,4.

20. A bag contains 5 white and 3 black balls. Find the probability of selecting a black ball.

Revision answers

1. −6.

2. 14.

3. 125.

4. 8.3×10^{-5}.

5. 8 days.

6. [[1,8],[4,4]].

7. $y = 3x + 2$.

8. $x \leq -4$.

9. 616 cm^2.

10. 63π cm^3.

11. 3 g/cm^3.

12. 72 km/h.

13. KSh 2,800.

14. 0.0088.

15. 3.

16. 36 km.

17. 9:16.

18. 10 cm.

19. 17.

20. 3/8.

Curriculum and use note

These are original teacher-prepared notes based on the Grade 9 Mathematics topic sequence in the annual scheme of work. They are intended to deepen understanding and provide practice, not to replace the KICD curriculum design, teacher instruction or approved learner's materials. The teacher may add local examples, additional exercises, projects and assessment tasks to match learner progress.

Reference: Kenya Institute of Curriculum Development, Grade Nine Mathematics Curriculum Design (2024): https://kicd.ac.ke/wp-content/uploads/2024/06/Mathematics-Grade-9-Design-Formatted-April-2024.pdf